The Attention Saturation Threshold: A Real-Time Coordination Stress Monitor for Global Equity Markets

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ABSTRACT

When investor attention crosses a structural boundary, price discovery does not slow - it switches. Three hours before crossing, realized volatility sits at 0.898 times baseline; one hour after, it reaches 1.877 times baseline across 22 global equity markets. No publicly documented macroprudential instrument detects this transition in real time (Section 2 benchmarks CoVaR, SRISK, and connectedness monitors directly). This paper identifies the Attention Saturation Threshold (AST), the critical attention density at which narrative synchronization displaces private-signal processing. Estimated on 22 equity markets across six continental regions over 2015–2025, the pooled-panel boundary is $\gamma^* = 1.5625$, associated with an 84.6 percent discrete volatility jump, a rise in cross-market correlation from 0.341 to 0.612, and an 11.54 percentage-point concentration of return variance into a single dominant global factor. The transition is sequential and independently identified at each node: belief compression precedes liquidity withdrawal, which precedes volatility amplification. The threshold specification dominates smooth alternatives by 504.6 BIC units. At the recommended deployment point (threshold = 0.921), GASI attains precision 0.563 and recall 0.722 on coordination episodes, while discriminating near-randomly on credit-driven stress including the 2023 SVB episode (AUC 0.51), confirming mechanism specificity. A 120-minute disclosure stagger reduces crossing probability by 62 percent under full market coverage at zero informational cost.

1. INTRODUCTION

Markets aggregate what investors know - until they do not. When attention density crosses a structural threshold, the aggregation mechanism breaks: prices stop reflecting private signals and start reflecting higher-order beliefs about what everyone else believes. This paper identifies the boundary where that transition occurs, builds a real-time monitor around it, and derives the policy intervention that delays it. The FSB (2023) Global Monitoring Report identifies synchronized investor behavior and liquidity mismatch as central surveillance priorities; the gap it documents is precisely the absence of a real-time trigger for this state change.

The defining empirical signature is the Silence Signature. Three hours before a global threshold crossing, belief heterogeneity compresses while realized volatility remains suppressed at 0.898 times baseline. Agents have already abandoned private signals and converged on the shared narrative before any price movement signals stress. By the time volatility-based monitors activate, the mechanism has completed its first two steps. One hour post-crossing, realized volatility reaches 1.877 times baseline across all 22 markets. This is a structural deficiency of realized-variance monitoring, not a timing accident.

The central empirical result is a pooled-panel threshold at $\gamma^* \approx 1.56$ - the attention density at which coordination incentives first dominate private-signal processing - associated with an 84.6 percent discrete volatility jump, cross-market correlation rising from 0.341 to 0.612, and network consolidation from 11 disconnected components to a single fully connected graph. Four independent identification procedures converge within a standard deviation of 0.019. The threshold is not a universal constant: it varies predictably with liquidity depth ($\partial\gamma^*/\partial D > 0$), so thinner markets cross earlier. Frozen at $\gamma^* = 1.5625$, the calibration carries forward into Q1 2026 monitoring without re-estimation - the governance stability criterion for supervisory deployment.

The mechanism is sequential and independently identified at each node. As IAI crosses γ^*: agents reduce private-signal weight and coordinate on the public narrative (Step 1: belief compression, $t = 4.58$); liquidity providers withdraw depth under synchronized one-sided order flow (Step 2: liquidity withdrawal, Lewbel IV $t = -2.90$); depleted depth converts marginal imbalances into outsized price revisions (Step 3: volatility amplification, pooled $t = 13.82$). Dual falsification - earnings-calendar and regulatory-disclosure IVs - rules out information-processing volume as the operative channel. The activating condition is narrative synchronization.

Identification rests on convergence across designs, not a single estimator. Profile likelihood, permutation ranking, and holdout diagnostics anchor $\gamma^* = 1.5625$. Within-country liquidity depth, instrumented by the 2020 COVID shock, confirms the structural comparative static from exogenous variation ($F = 30.08$, second-stage $t = 5.21$). Official exchange cross-validation establishes data fidelity independent of any single vendor.

The policy output is GASI: a mechanism-specific daily trigger that activates on coordination episodes, abstains on SVB-type credit stress, and maps to a pre-committed intervention margin under frozen governance. A 120-minute disclosure stagger reduces simulated crossing probability by 62 percent without altering information content - the second-best timing policy from the Bergemann–Morris (2019) information-design framework.

Section 2 positions the paper. Sections 3–4 develop data and theory. Sections 5–6 identify the threshold and mechanism. Section 7 documents cross-market heterogeneity. Sections 8–9 report robustness and policy translation. Section 10 concludes.

2. RELATED LITERATURE

This paper contributes to three strands - attention pricing, liquidity amplification, and systemic connectedness - and is additive to each by identifying the endogenous state variable, attention density at the coordination boundary, that determines when each framework's amplification logic activates.

Attention-based asset-pricing models estimate smooth relationships between attention proxies and returns or volatility (Da, Engelberg, and Gao, 2011; Andrei and Hasler, 2015; Bybee, Kelly, and Manela, 2024). The contribution here is a state-dependent extension: linear mapping approximates well below the boundary; coordination dynamics govern above it. The Bybee, Kelly, and Manela (2024) factor correctly prices the low-saturation regime comprising approximately 91 percent of market-days; the remaining 9 percent constitute a qualitatively distinct coordination regime in which the aggregation mechanism changes form.

The transmission channel connects to liquidity and intermediary balance-sheet work (Brunnermeier and Pedersen, 2009; Adrian and Shin, 2010, 2014). Those frameworks require risk-bearing capacity to contract when measured risk rises and funding tightens. This paper provides the endogenous activation margin: crossing states are precisely the conditions under which one-sided order flow, margin sensitivity, and depth withdrawal become jointly reinforcing. The contribution is a trigger condition, measurable in real time, for when liquidity spirals become likely at market-wide scale.

Hogen, Kasai, and Shinozaki (2025), published in this journal, document the global structural change in shock transmission driven by rising NBFI importance, quantifying propagation once co-movement is established. GASI and connectedness frameworks operate at different points in the stress timeline and are designed for sequential deployment: GASI detects the pre-propagation coordination state before rolling-window connectedness measures move; connectedness monitors then quantify transmission structure. The two tools are complementary surveillance layers, not competing alternatives.

The theoretical foundations are Morris and Shin (2002), Vives (2014), Goldstein and Pauzner (2005), and Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015). The empirical design follows Bai and Perron (1998) and Hansen (1999), augmented with convergent causal checks, dual falsifications, and frozen holdout governance. Prior work explains how risk transmits; this paper identifies when the system enters the transmission-prone state.

A practical distinction is timing resolution. Rolling-window diagnostics are informative retrospectively but slow for intervention. Combining daily threshold detection with intraday event anatomy (Section 6.4) frames policy logic in transition-time rather than quarter-time terms - the relevant supervisory horizon when crossings are fast and disclosure timing can still alter outcomes.

3. DATA AND EMPIRICAL INFRASTRUCTURE

The panel is constructed to identify a global coordination mechanism, not a US-specific episode or a crisis-period artifact. It spans 22 markets across six continental regions and estimates the same threshold effect in both event-window and unfiltered panels. In the unfiltered daily panel ($N = 56{,}560$), the threshold coefficient is positive and strongly significant ($\beta = 0.0060$, $t = 6.41$, $p < 0.001$); event-window conditioning sharpens mechanism interpretation without inflating the result. The event-window panel is the primary identification vehicle; the unfiltered panel anchors sign and magnitude stability.[1]

3.1 Universe and Coverage

The panel covers 22 major equity indices over 2015-01-15 to 2025-12-28: Australia (AS51), Belgium (BEL20), Brazil (BVSP), Canada (GSPTSE), China (SSEC), Eurozone

[1] The unfiltered daily panel uses $N = 56{,}560$ country-day observations across 22 markets over 2015-2025 under pooled OLS without event-window conditioning. The event-window panel ($N = 4{,}109$) applies Driscoll-Kraay dependence correction at L=10 lags and restricts attention to event-proximate windows for sharper mechanism identification. Both panels yield positive, strongly significant threshold coefficients; the difference in coefficient magnitudes reflects conditioning scope and error-structure correction, not contradiction. See Online Appendix Section G for filter-concordance diagnostics.

(STOXX50E), France (FCHI), Germany (GDAXI), Hong Kong (HSI), India (BSESN), Indonesia (JKSE), Japan (N225), Malaysia (KLSE), Mexico (MXX), Netherlands (AEX), New Zealand (NZ50), Singapore (STI), South Korea (KS11), Spain (IBEX), Taiwan (TWII), United Kingdom (FTSE), and United States (GSPC). Local index labels are reported in-text for readability, while the replication pipeline uses Yahoo ticker tags (for example, Australia ^AXJO, Belgium ^BFX, China 000001.SS, and India ^NSEI). Returns are computed in local currency; USD conversion robustness is reported separately. Taiwan Roll spread is constructed from ticker-level return reversals due to tick-data limits, and core liquidity-chain inference is unchanged when Taiwan is excluded. Roll (1984) spread computation requires negative serial return covariance, a condition that is not guaranteed in daily index returns. For completeness, we report computability rates by market in Online Appendix Section H: across the 22-market panel, Roll spread is computable on a mean of 73.4 percent of country-days, ranging from 64.0 percent for markets with closing auction mechanisms that introduce positive autocorrelation (notably Japan and Singapore) to 84.6 percent for markets with continuous trading close procedures. Non-computable observations are assigned the market's rolling 30-day median spread. As a robustness check, all liquidity-chain results are reproduced using the Corwin-Schultz (2012) high-low OHLC spread estimator, which does not require negative serial covariance and is available for all observations; results are reported in Online Appendix Section D6 and are qualitatively identical to the Roll-based primary estimates.

All daily price and volume data are sourced from Yahoo Finance (2015-2025) via yfinance. Cross-validation against official exchange-published close and volume series in five representative markets (US, UK, DE, JP, BR) shows mean absolute price deviation below 0.01 percent and mean absolute volume deviation below 1.2 percent (Online Appendix Table S58A). Hourly intraday series for Section 6.4 use one-hour bars with staggered market availability from 2017 to 2019; pre-coverage intervals are excluded. Intraday timestamps are localized by market timezone and converted to UTC before session clustering; U.S. session hours are defined as 14:00-21:00 UTC in the trigger-architecture diagnostics. Data quality controls include drift flags, manual reconciliation for flagged observations, and robustness to Taiwan exclusion (Online Appendix Section H). Q1 2026 monitoring results are treated as post-sample prospective validation and are not part of the estimation panel ending 2025-12-28.

Direct control for Global Economic Policy Uncertainty leaves the threshold effect unchanged

($\beta = 0.0004$, $p = 0.001$).[2]

Figure 1 reports the threshold discovery architecture: Panel A (profile-LR surface), Panel B (permutation null), and Panel C (leave-one-out stability band).

Figure 1: Threshold Discovery Architecture. Panel A shows the profile likelihood ratio surface over 81 candidate values; Panel B shows the permutation null distribution; Panel C shows the leave-one-out stability band. The profile LR peak at $\gamma^* = 1.5625$ ranks at the 98.43rd permutation percentile, supporting threshold identification under the reported permutation design. Source: Author calculations.

Table 1: Primary Data Sources and Quality Protocols.

Region	N markets	Date range	Markets	Primary Source	Protocols
Americas	4	2015-01-15 to 2025-12-28	US, CA, BR, MX	Yahoo Finance (yfinance API)	Closing-price drift check < 0.05%
Europe	8	2015-01-15 to 2025-12-28	UK, DE, FR, EU, etc.	Yahoo Finance (yfinance API)	Closing-price harmonization applied using market-specific UTC offsets
Asia-Pac	10	2015-01-15 to 2025-12-28	CN, JP, HK, TW, etc.	Yahoo Finance (yfinance API)	Pre-2018 volume drift filter; closing-price drift check < 0.05%

Note: All data are synchronized to UTC using market-specific closing offsets. Volume series exceeding 10% daily drift are flagged for manual reconciliation.

Event-window conditioning uses a pre-coded multi-country corpus of policy, macro, geopolitical, market-structure, and regulatory events. The corpus is used only for conditioning and

[2] A Hausman-type endogeneity test for the GEPU regressor yields $p = 0.312$, supporting the exogeneity assumption; results are robust to lagged GEPU specifications. The GEPU-controlled row in Table 3 is interpreted as a conservative over-control bound because GEPU co-moves with synchronization states near the threshold; see Table 3 note for full reconciliation.

announcement splits; threshold-location estimation uses the full unfiltered panel. Category counts and source mapping are reported in Online Appendix Section H.

3.2 Sentiment Measurement and Multi-Model Robustness

The coordination failure at γ^* is invariant across the main-text NLP architectures (GPT-4 and VADER), with supplementary FinBERT and bag-of-words checks reported in the replication outputs; cross-model dispersion is minimal (SD = 0.000018). Volume-based attention proxies contain a liquidity component that residualization only partially addresses; future work with direct retail order-flow data would sharpen this distinction.

All primary identification variables are derived entirely from Yahoo Finance price and volume series. Primary inference is therefore independent of the sentiment and event corpus used in robustness specifications.

3.3 External Validation

External validation is anchored to exchange-level cross-checks and independent robustness diagnostics in the appendix. The purpose is to verify that the synchronization signal is not a single-vendor artifact while preserving full reproducibility within the submitted package.

3.4 The Investor Attention Index

The primary explanatory variable is the Investor Attention Index:

$$\mathrm{IAI}_{i,t} = V_{i,t} \; / \; \mathrm{MA}_{30}(V_{i,t})$$

where V represents daily trading volume and MA_{30} denotes the 30-day trailing moving average. A ratio formulation is preferred to absolute volatility scaling because it normalizes baseline regime differences across markets, decades, and liquidity structures. This scale-free ratio enables direct cross-market comparison. Random matrix decomposition identifies a single dominant global attention factor: the leading eigenvalue of 3.546 exceeds the Marchenko-Pastur upper bound (1.227) by 2.319. The factor is recovered from the data rather than imposed by construction.

Using the external validation in Section 3.3, the threshold effect remains significant after residualization against lagged liquidity, news intensity, and macro announcements (p = 0.003).

These diagnostics jointly separate attention synchronization from pure liquidity mechanics: the residualized threshold effect ($p = 0.003$) and the earnings-IV null continue to support the same channel interpretation.

The primary outcome variable is the daily change in 30-day realized volatility:

$$\Delta\sigma_{i,t} = \sigma_{i,t} - \sigma_{i,t-1}, \quad \sigma_{i,t} = \sqrt{252} \cdot \sqrt{\frac{1}{29}\sum_{j=1}^{30}(r_{i,t-j} - \bar{r})^2}$$

The 30-day window balances noise reduction with regime responsiveness; robustness to 20-day and 60-day alternatives is confirmed in Section 8. Results are stable to the Parkinson (1980) range-based estimator ($\beta = 0.0005$, $p < 0.01$).

Replication package contents include scripts for GASI construction, threshold scanning, inference diagnostics, policy Monte Carlo, and figure/table reproduction with fixed software environment metadata and random seeds.

Table 2: Panel Summary Statistics (Daily Estimation Panel, N=4,109).

Statistic	Full Sample	DM Subsample	EM Subsample
Mean IAI	1.092725	1.12977	1.033124
Mean VolDelta	0.000378	0.000472	0.000226
Mean VolBaseline	0.007466	0.007405	0.007564
High-Regime Share (IAI>=1.5625)	8.96%	10.93%	5.78%
Countries	22	14	8

Note: EM markets cross less often because estimated threshold levels are lower; the fragility result concerns amplification conditional on crossing. Intraday anatomy uses all available panel-hour observations across the 22-market panel under available hourly coverage windows. Inferential tables use the 4,109 daily rows. Source: Author calculations.

Table 3: Primary Baseline Inference Stack.

Specification	Beta(IAI threshold jump)	p-value	Inference
Driscoll-Kraay FE (L=10)	0.001307	<0.001	Primary (event-window conditioned)
Wild Cluster Bootstrap	0.000599	<0.001	Primary reinforcement
Fama-MacBeth (Monthly CS + NW)	0.002236	<0.001	Independent cross-sectional anchor
Country x Month Clustering	0.001285	0.027	Conservative bound
Country x Year Clustering	0.001285	0.037	Conservative bound
Conley Spatial HAC (8,000km)	0.001300	0.013	Conservative context
GEPU-controlled specification	0.000400	0.001	Macro-uncertainty control check

Note: Source: Author calculations. The GEPU-controlled row adds GEPU directly to the event-window conditioned baseline with the same country and time fixed effects and dependence-robust inference stack. Because GEPU partly co-moves with synchronization states near crossings, attenuation in this row is interpreted as a conservative over-control bound rather than a replacement central estimate. The sample and dependent variable are unchanged relative to the baseline row.[3]

Across all seven inference treatments in Table 3, the threshold-jump coefficient remains positive and statistically significant; inference-engine choice affects precision but not sign or economic interpretation. The attenuation in the GEPU-controlled row reflects partial overlap between macro-uncertainty proxies and synchronization states near threshold, so this specification is interpreted as a conservative over-control bound rather than the central causal estimate. In a compact fast-spec diagnostic subset excluding the most conservative clustering variants, the range is 0.002241 to 0.003392 (cross-specification SD = 0.000578), all positive and significant.

[3]The Driscoll-Kraay primary estimate of 0.001307 and the unfiltered pooled OLS of 0.0060 (Section 3) are not directly comparable. The Driscoll-Kraay estimate uses event-window conditioning with Driscoll-Kraay dependence correction at L=10 lags; the 0.0060 estimate uses the full 56,560-observation unfiltered daily panel under pooled OLS without event conditioning. Both are positive and strongly significant; the difference in magnitude reflects conditioning scope and error-structure correction, not a contradiction.

4. THEORETICAL FRAMEWORK AND STRUCTURAL MAPPING

4.1 Coordination Structure and Threshold Definition

Each agent solves a dual problem: forecast the fundamental value y_t, and coordinate with the market consensus. Agents observe a private signal $s_{i,t} = y_t + \epsilon_{i,t}$ and a public narrative signal $z_t = y_t + \eta_t$. Following Morris and Shin (2002), payoffs penalize both forecast error and departure from the market action:

$$U_i = -(a_{i,t} - y_t)^2 - \kappa_t \int (a_{i,t} - a_{j,t})^2 dj.$$

As κ_t rises, acting on heterogeneous private signals becomes costly, and the public narrative dominates equilibrium actions. The Attention Saturation Threshold (AST) is the density γ^* at which coordination weight first becomes dominant. Under heterogeneous thresholds and logistic aggregation,

$$\kappa_t = \frac{1}{1 + e^{-\alpha(IAI_t - \gamma^*)}},$$

producing a unique transition boundary and an inflection in sensitivity exactly at γ^*.

4.2 Volatility Discontinuity and Mechanism Logic

As κ_t approaches unity, agents progressively suppress private signals and anchor on the common public narrative. The chain rule delivers the central theoretical result: volatility sensitivity to attention is maximized precisely at γ^*, not sustained above it, producing a sharp discontinuity rather than a smooth inflection:

$$\Delta\sigma^2\big|_{\gamma^*} = \int_{\gamma^*-\epsilon}^{\gamma^*+\epsilon} \frac{\partial \mathrm{Var}(a_t)}{\partial \kappa} \frac{\partial \kappa}{\partial IAI} \, dIAI.$$

The empirical implication is a regime boundary, not a slope change: below-threshold dynamics are linear and well-approximated by standard attention models; above-threshold dynamics are coordination-dominated and discontinuous. The BIC gap of 504.6 units in favor of the threshold specification, over 50 times the conventional decisiveness threshold of 10 BIC units, confirms that the data prefer a boundary to a smooth alternative.

4.3 Endogenous Liquidity and Hysteresis

Synchronized order flow creates state-dependent depth withdrawal. The reduced-form depth response,

$$D(t) = D_0(1 - \phi\, m(t)^+), \quad m(t)^+ = \max(\kappa_t - 1, 0),$$

yields a state-dependent effective threshold:

$$\gamma^*(t) = \frac{D_0(1 - \phi\, m(t)^+)\tau}{\lambda}.$$

As coordination mass rises, order-flow one-sidedness becomes persistent: inventory risk rises, quote shading widens, depth contracts, and the marginal distance to the next crossing falls. The result is hysteresis: entry into the high-synchronization regime requires a larger shock than persistence within it; exit requires lower attention density and re-expansion of both depth and risk-bearing capacity. The mechanism parallels Brunnermeier and Pedersen (2009) and Adrian and Shin (2014), with one distinction: the ignition variable is narrative synchronization, not prior balance-sheet impairment. The sign of ϕ is identified through mechanism ordering and post-crossing depth behavior in Section 6; level identification requires dealer-inventory data unavailable at country-day aggregation.

4.4 Testable Predictions

The framework yields four empirical predictions tested in Sections 5-8:

1. Belief compression rises discontinuously at γ^*.
2. Liquidity withdrawal is causal and follows attention coordination.
3. Volatility amplification scales with illiquidity after crossing.
4. Cross-market topology consolidates via a common synchronization component above threshold.

These predictions are evaluated independently using distinct identification designs and summarized in Online Appendix Section A1.8.

4.5 Liquidity Architecture Comparative Statics

Define $\kappa = \kappa(IAI, D)$ with $\partial\kappa/\partial IAI > 0$ and threshold condition $\kappa(\gamma^*, D) = 1$. If deeper liquidity lowers coordination incentives ($\partial\kappa/\partial D < 0$), then

$$\frac{\partial\gamma^*}{\partial D} = -\frac{\partial\kappa/\partial D}{\partial\kappa/\partial IAI} > 0.$$

Thus thinner-depth markets should cross at lower attention density and amplify more strongly post-crossing.

4.6 Structural Mapping and Identification Boundary

The structural mapping is $\gamma^* = D\tau/\lambda$. Depth D is identified from within-country exogenous variation, while λ is not point-identified at country-day aggregation. This affects level scaling, not the sign of the comparative static $\partial\gamma^*/\partial D > 0$; Section 10 discusses the tick-level dealer-inventory data required for primitive-level pinning.

4.7 Within-Country Liquidity-Depth Identification via the COVID Shock

To isolate the comparative static $\partial\gamma^*/\partial D > 0$ from cross-country proxy noise, we estimate a within-country annual panel using a COVID-shock instrument for liquidity depth. The second-stage dependent variable is the annual rolling threshold location estimated by profile likelihood over non-overlapping 252-trading-day windows, producing one observation per country-year. The estimation sample contains $N = 193$ country-year observations across 22 markets, with lagged depth proxied by inverse Roll spread. The first stage is strong: the 2020 binary shock indicator (equal to one during calendar year 2020 and zero otherwise) loads materially on the depth proxy (coefficient on COVID = 9.152879, $t = 5.484611$, $p = 4.143799 \times 10^{-8}$; first-stage $F = 30.08096$). The second stage maps predicted depth into annual threshold location and yields a positive, precisely estimated effect (coefficient on $\widehat{Depth} = 0.005009229$, $t = 5.207017$, $p = 1.919009 \times 10^{-7}$). This is the central within-country quasi-natural-experiment design in the paper: it directly instruments liquidity depth and recovers the structural comparative static from exogenous variation rather than cross-country levels.

Table 4: Within-Country Structural Identification, COVID Liquidity Shock IV.

Stage	Specification	Coefficient	t-stat	p-value	F-stat
First stage	COVID dummy → Depth (inverse Roll, lagged)	9.152879	5.484611	4.143799×10^{-8}	30.08096
Second stage	Predicted Depth → γ^*	0.005009229	5.207017	1.919009×10^{-7}	-

Note: Second-stage dependent variable is annual rolling threshold location γ^* from profile likelihood on 252-day windows. Sample size is $N = 193$ country-year observations across 22 markets. Market and year fixed effects are included.

4.8 Cross-Asset Corollary and Information-Design Policy Mapping

Intervention instruments differ by mechanism node. Below threshold with rising synchronization pressure, information-design tools that alter disclosure timing are first-best: they target the common-signal channel without suppressing information content. Above threshold with active depth withdrawal, timing tools may be insufficient and should be paired with market-function tools that preserve liquidity provision capacity. Using one template across pre-crossing and post-crossing states is the policy mistake this framing prevents.

Disclosure staggering is coherent in welfare terms. In a Morris-Shin setting, welfare losses arise from excessive common-signal weighting even when information is accurate. Staggering reduces synchronous reaction intensity while preserving the total informational set, representing a re-timing rule rather than censorship. Policy success should therefore be measured on crossing probability and Expected Shortfall, not average volatility; a successful timing intervention may leave unconditional volatility unchanged while materially reducing tail amplification frequency.

Two corollaries follow. First, initiation is architecture-dependent; terminal amplification is architecture-invariant. Pre-crossing calm concentrates in equities, where market-making inventory dynamics produce sharper depth withdrawal, while Step-3 amplification replicates across commodities and crypto once liquidity buffers compress (Online Appendix Table S68). Second, the policy intervention is a timing problem, not an information-suppression problem. In the Bergemann and Morris (2019) framework, the planner chooses the temporal structure of signal revelation. The 120-minute stagger applies this directly: de-synchronizing high-salience public signals near the boundary lowers crossing probability without reducing total informational content.

Full theorem statements and proofs are in Online Appendix Section A.

4.9 Social Planner Threshold Theorem and Second-Best Timing Policy

The planner faces constrained mechanism design: private-signal precision is fixed, strategic complementarities cannot be removed, and heterogeneous beliefs cannot be forced. The implementable margin is timing architecture, reducing synchronous updating intensity near the boundary rather than altering fundamentals. This matches the realistic toolkit of supervisors, who hold communication and sequencing authority but do not control private-information production.

Transition probability is locally steep near γ^*, so small boundary-region reductions in synchronization speed generate disproportionately large declines in tail-state incidence; this is why the calibration reports equivalent-threshold uplift alongside crossing-probability reduction rather than unconditional variance summaries. The paper claims a bounded, implementable welfare improvement at this operational margin, not global welfare maximization.

Formally: let welfare loss per crossing be $L > 0$, crossing probability $P_{\text{cross}}(\gamma)$, and liquidity provision cost convex $C(D)$. With $\gamma^* = D\tau/\lambda$:

$$\max_D W(D) = -L \cdot P_{\text{cross}}(\gamma^*(D)) - C(D), \qquad \text{FOC: } L\left|\frac{\partial P_{\text{cross}}}{\partial D}\right| = C'(D).$$

This defines D^* and $\gamma^*_{\text{opt}} = D^*\tau/\lambda$. If $D_{\text{actual}} < D^*$, markets cross too frequently relative to the social optimum. The 120-minute stagger does not change D directly; it lowers synchronization density and therefore P_{cross} at fixed D.

Proposition 1 (Socially Optimal Threshold and Second-Best Timing). Under $\gamma^* = D\tau/\lambda$, convex $C(D)$, and crossing-loss schedule $L \cdot P_{\text{cross}}(\gamma)$, the planner chooses D^* to balance marginal crossing-loss reduction against marginal liquidity cost. If $D_{\text{actual}} < D^*$, crossings are excessive. A timing-stagger reducing crossing probability by 62% implements a second-best effective-threshold increase $\Delta\gamma_{\text{equiv}}$, defined by $P_{\text{cross}}(\gamma^*_{\text{actual}} + \Delta\gamma_{\text{equiv}}) = 0.38 \cdot P_{\text{cross}}(\gamma^*_{\text{actual}})$. Full derivation in Online Appendix Section OA-18.

Empirically, the 62 percent P_{cross} reduction implies $\Delta\gamma_{\text{equiv}} \approx 0.09$, an effective uplift from 1.5625 to approximately 1.65, representing illustrative policy arithmetic under the baseline calibration.

Figure 2: Social Planner Threshold Theorem Extension. Panel A maps empirical crossing probability to threshold level and marks the policy-equivalent threshold after the 120-minute stagger. Panel B reports the implied second-best welfare gain under normalized crossing-loss units. Takeaway: timing coordination shifts the system toward the social optimum by raising the effective threshold without suppressing information. Source: Author calculations.

5. THRESHOLD IDENTIFICATION

5.1 Convergent Identification Stack and Frozen Calibration

Threshold framing note. The estimated value $\gamma^* = 1.5625$ is the pooled-panel profile-likelihood peak across the full 22-market panel and training sample. It is the governance anchor for GASI deployment. It is not claimed to be a universal constant identical across all markets. The structural comparative static $\partial\gamma^*/\partial D > 0$ implies that countries with thinner liquidity architecture have lower operational thresholds: the EM subsample profile-likelihood peak sits near 1.18 (profile-likelihood CI reported in Online Appendix Section D2), while DM markets cluster above 1.50. Country-specific threshold variation is documented in Section 7 and Online Appendix Section D2; $\gamma^* = 1.5625$ should therefore be read as the cross-market coordination baseline, with local depth architecture governing deviations from it.

An important alternative explanation is that any high-attention episode, regardless of source, triggers coordination failure. The earnings-calendar design provides evidence inconsistent with that interpretation. Scheduled filing density generates strong first-stage attention variation ($F = 25.65$), so earnings traffic can mechanically push IAI high. The second stage is null ($\beta = -0.005$, $p = 0.693$): those crossings do not produce saturation volatility. The operative variable is narrative synchronization, not information-processing volume. A potential concern is that the exclusion restriction for the earnings-calendar instrument is violated if earnings filing density affects volatility directly through price discovery rather than only through attention volume. We address this in three ways. First, the instrument is the aggregate density of filing activity across all firms in the full 22-market panel simultaneously, not any single firm's earnings release; at this aggregation level, firm-specific price-discovery effects are orthogonal to the instrument by diversification. Second, the null second stage is itself evidence against direct price-discovery contamination: if earnings density were generating volatility through information channels directly, we would observe a positive second stage, not a null. Third, the same null second-stage pattern appears in the independent regulatory-disclosure IV, which uses mandatory filing deadlines rather than earnings information; these deadlines are procedural calendar events with no price-discovery content, providing a cleaner exclusion. Convergent nulls across two instruments with different exclusion concerns jointly rule out information-processing volume as the operative channel.

Regulatory-disclosure IV as independent falsification. A second falsification design uses cross-country variation in mandatory disclosure windows. Filing deadline dates are sourced from each country's primary securities regulator (SEC EDGAR, FCA, AMF, BaFin, SEBI, and equivalents across all 22 markets) and constructed as a binary indicator active in the five trading days preceding each mandated deadline. For each country-day, the instrument equals one in the five trading days preceding a mandated filing deadline; global disclosure density is the daily count of countries simultaneously in-window. On the full 2015-2025 unfiltered 22-market equity panel, the first stage is strong ($\beta = 0.0536$, $t = 6.56$, $p = 5.26 \times 10^{-11}$, $F = 43.08$) and the second stage is null ($\beta = -0.000483$, $p = 0.825$). Regulatory-deadline attention therefore behaves like earnings-calendar attention: it raises processing volume but does not trigger saturation volatility. Two independent scheduled-attention instruments produce the same pattern, jointly ruling out information-processing volume as the operative channel.

The identification challenge is explicit: attention, liquidity, and volatility co-move and become jointly endogenous near the threshold.

Identification rests on convergence across four independent procedures that recover $\gamma^* = 1.5625$ (95% CI: $[1.551, 1.574]$ via profile likelihood with wild-cluster bootstrap support) within a standard deviation of 0.019 across inferential engines. All panel regressions use Driscoll-Kraay (1998) standard errors at L=10 lags, with alternative inference stacks reported in Table 3.

Threshold discovery is training-partition anchored: γ^* is identified on 2015-2020, while inferential tests, heterogeneity estimates, and policy simulations are evaluated on the 2021-2025 holdout under frozen governance. No threshold parameter crosses from training to holdout. IPW and entropy balancing confirm covariate balance; the entropy-balanced estimate of 0.00147 is the conservative lower bound (Online Appendix Section G). For fear-proxy falsification, the low-VIX attention subsample is defined as IAI above 1.5625 with VIX below the full-sample median of 17.3 ($N = 233$), used in Table 9 Panel D.

The placebo on baseline volatility yields a profile likelihood ratio 852 times smaller than the attention threshold; the result is not a disguised volatility effect.

Structural placebos re-estimate the procedure on randomized country-time assignments, a non-equity panel, and a US-only panel. None produces a stable threshold under the same discovery protocol. The $\gamma^* \approx 1.56$ result reflects synchronized global equity coordination, not a quirk of the estimation method (Online Appendix Table S63).

A round-number alternative at $\gamma = 2.0$ would classify a substantially smaller fraction of observations as high-regime. The data do not support $\gamma = 2.0$ under three independent checks. The high-regime sample falls 54 percent. Bootstrap concentration centers at 1.513. Structural calibration inflects at 1.5714. The threshold is $\gamma = 1.5625$.

$\gamma = 2.0$ rejection check	Evidence
High-regime support	Falls 54 percent versus $\gamma^* = 1.5625$ split
Bootstrap concentration	Centers at 1.513, not 2.0
Structural calibration	Inflects at 1.5714, not 2.0

The threshold recalibrates with market liquidity conditions across the sample period. Rolling annual estimates range from 1.464 in 2016 to 1.518 in 2024, with a V-shaped pattern during the pandemic period (1.279 in 2020, recovering to 1.498 by 2022) reflecting structural compression of liquidity depth during COVID stress and subsequent normalization. This recalibration is consistent with the theoretical prediction $\gamma^* = D\tau/\lambda$: periods of thinner liquidity depth generate lower operational thresholds, as confirmed by the EM versus DM comparison in Section 7. The primary estimate of 1.5625 represents the full-sample profile-likelihood peak and is the appropriate frozen calibration for out-of-sample monitoring; rolling estimates characterize time variation in liquidity architecture rather than a different structural parameter.

5.2 Endogenous Liquidity and the Feedback Loop

The within-country COVID-shock IV is the central quasi-natural experiment for the structural comparative static. It instruments liquidity depth variation directly and recovers the predicted sign of $\partial\gamma^*/\partial D$ in a fixed-effects panel, rather than inferring structural direction from cross-country levels.

Identification via Lewbel (2012). The Lewbel heteroscedasticity-based IV addresses endogeneity in the liquidity-withdrawal channel without requiring an external instrument. The instrument takes the form $Z_i = (X_i - \bar{X})\hat{e}_i$, where X_i are demeaned exogenous controls and $\hat{e}_i$ are first-stage residuals. Validity requires that the heteroscedasticity driving the instrument exists in the sub-threshold regime independently of post-crossing coordination effects. The sub-threshold Breusch-Pagan test ($N = 3{,}741$, LM $= 397.88$, $p = 3.99 \times 10^{-87}$) confirms this structure pre-dates any crossing, establishing non-circularity. The IV estimate ($t = -2.90$; first-stage $F = 126.50$) confirms the causal arrow runs from attention to liquidity even after accounting for the structural feedback loop.

The Breusch-Pagan test rejects homoscedasticity in both regime partitions ($p_{\text{sub}} = 8.33 \times 10^{-99}$; $p_{\text{high}} = 4.48 \times 10^{-4}$); the identifying moments are structural, not sampling artifacts.

The earnings IV falsification sharpens identification. Using earnings filing density as instrument (first-stage $F = 25.65$), the 2SLS second stage is null ($\beta = -0.005$, $p = 0.760$). Scheduled earnings-driven attention does not trigger saturation volatility. The operative channel is narrative synchronization, not information-processing volume.

5.3 Regression Discontinuity: Reported for Transparency, Not for Identification

RDD is structurally inappropriate here and is reported for completeness only. The running variable is endogenous attention density, not an exogenous administrative cutoff. Local exchangeability is violated by construction: volume-driven crossings are systematically related to the outcome, so the identifying condition for causal local-linear RDD fails at the design stage, not at the estimation stage. Identification in this paper relies exclusively on the convergent stack in Section 5. The CCT local-linear estimate at $\gamma^* = 1.5625$ is null ($p = 0.232$), at a bandwidth of $h = 0.031$ with $N = 58$ boundary observations. This null is the architecturally expected result and carries no inferential weight against the threshold architecture. A conventional local-randomization sample near $N \approx 200$ would require $h \approx 0.10$, a neighborhood wide enough to pool heterogeneous liquidity regimes and destroy local comparability.

5.4 News-Independent Attention Residuals

The index is residualized against news-event counts, macroeconomic announcements, VIX level, and lagged volatility. The news-independent attention residual remains highly significant ($\beta = 0.0003$, $t = 3.18$, $p = 0.0015$).

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6. TRANSMISSION MECHANISM AND SEQUENTIAL CHAIN

6.1 Step 1: Attention Saturation Triggers Coordination

The mechanism initiates with belief compression. As IAI crosses γ^*, the rolling cross-sectional standard deviation of daily returns contracts sharply ($t = 4.58$, $p < 0.001$), the market-level signature of reduced private-signal weighting as agents coordinate on the shared public narrative. Within-market sentiment dispersion contracts in the same direction (quintile slope $p = 0.037$).

The causal direction is supported by dual falsification. Neither lagged dispersion predicts high IAI ($p = 0.280$) nor does future high IAI predict current dispersion ($p = 0.331$), ruling out reverse causation. An independent analyst forecast-dispersion measure contracts above the threshold ($p < 0.05$; Online Appendix Table S66), confirming the belief-compression pattern in an external data source. IAI Granger-causes realized volatility in an AIC-selected lag structure (optimal lag $= 2$; $p < 0.001$).

6.2 Step 2: Coordination Triggers Liquidity Withdrawal

Synchronized order flow imposes one-sided demand on market makers structurally limited by capital constraints. The mechanism operates through both arms of the Brunnermeier-Pedersen (2009) liquidity spiral simultaneously: the coordination crossing generates a loss spiral via one-sided price pressure on existing inventory, and a margin spiral via elevated volatility inducing tighter collateral requirements. Step 2 is significant under both OLS ($t = -3.55$) and Lewbel IV ($t = -2.90$, $p = 0.004$).[4] The larger-magnitude IV estimate relative to OLS is consistent with attenuation bias correction: measurement error in volume-based coordination proxies biases OLS toward zero; the Lewbel instrument recovers the larger structural coefficient by purging this attenuation. Using the Step-2 coefficient ratio from the mechanism stack (DR-DiD 0.001832 versus Lewbel IV 0.011083), the implied attenuation factor is approximately 0.165; under a classical errors-in-variables approximation this maps to a signal-to-noise ratio near 0.20 for the volume-based proxy, consistent with treating IV as the structural coefficient anchor rather than OLS. As established in Section 5.2, the identifying variation is non-circular, exploiting heteroscedasticity documented exclusively in the sub-threshold subsample (Breusch-Pagan LM $= 397.88$, $p = 3.99\times10^{-87}$), which predates any crossing. Microstructure diagnostics confirm the mechanism: the Roll effective spread

[4] Lewbel diagnostics: first-stage F $= 126.50$; second-stage coefficient and sign are reported in the main Step-2 table.

rises 1.83 times above threshold ($p < 0.001$) and the price-reversal sign flips from positive to negative ($p = 0.011$) - the canonical adverse-selection signature of withdrawn depth provision. The AST is the attention density at which price impact becomes discontinuous.

6.3 Step 3: Illiquidity Amplifies Realized Volatility

With liquidity buffers withdrawn, the transmission chain reaches its terminal node: marginal order-flow imbalances generate outsized price revisions against a background of depleted market depth. The asset-specific replication is consistent across classes. In equity markets, the Roll spread (an illiquidity measure whereby higher spread indicates lower depth) predicts stronger amplification ($t = -2.31$, $p = 0.021$), exactly the depleted-depth direction. Commodities replicate the terminal channel via Amihud illiquidity ($t = 5.14$, $p = 2.8 \times 10^{-7}$). Cryptocurrency replicates via Corwin-Schultz spreads ($\beta = 0.186$, $t = 10.24$). Pooled across all three asset classes, the aggregate Step-3 effect is $t = 13.82$.

Controlling for the liquidity channel reduces the direct threshold coefficient by 37.5 percent. This implies that liquidity withdrawal accounts for an estimated one-third of the reduced-form threshold effect under a linear mediation decomposition (point estimate 37.5%), with the remainder transmitted through direct and parallel coordination channels. This decomposition is reported under linear mediation assumptions; under nonlinear coordination dynamics, the share should be interpreted as indicative rather than structurally exact.

Hysteresis and Recovery Deficit. The recovery deficit corroborates the liquidity interpretation and constitutes a distinct empirical contribution. Across 601 documented spell transitions (304 distinct spells), the recovery deficit is -0.002859 ($t = -5.849$, $p < 0.001$), confirming that post-exit normalization is structurally slower than pre-entry buildup. Median spell duration is one trading day, the 90th percentile is two days, and the maximum is four days; 83.2 percent of spells are single-day events. The asymmetry is architecturally predicted: entry is driven by synchronization pressure accumulating against a liquidity backstop; exit requires dealer inventory to be rebuilt and belief dispersion to re-establish, both of which depend on supply-side recovery time rather than demand-side reversal. The recovery deficit is therefore a depth-architecture prediction, not a persistence artifact.

Table 5: Condensed Three-Step Mechanism Summary

Mechanism block	Anchor statistic	Interpretation
Step 1: Belief compression	$t = 4.58$, $p < 0.001$	Dispersion contracts at crossing
Step 2: Liquidity withdrawal (OLS)	$t = -3.55$	Depth withdrawal follows coordination
Step 2: Liquidity withdrawal (Lewbel IV)	$t = -2.90$, $p = 0.004$	Causal direction robust to feedback
Step 3: Amplification (pooled cross-asset)	$t = 13.82$	Terminal amplification generalizes across assets
Mediation reduction	37.5 percent	Liquidity is a first-order transmission channel

Figure 3: Sequential Transmission and Sentiment Decoupling. Top panels: (A) standardized path coefficients for the three-step chain; (B) falsification scoreboard contrasting causal paths and placebos. Bottom panel (C): log-scale decoupling ratio (ASDR), showing attention-threshold effects dominate linear sentiment slopes. Takeaway: the three-step chain is jointly coherent in magnitude and sign, while placebo paths fail, supporting a coordination mechanism rather than a sentiment-level effect. Source: Author calculations.

Result 1 (The Silence Signature). In the three hours preceding threshold crossing, hourly realized volatility is systematically suppressed at 0.898 times baseline with no detectable upward drift, while belief heterogeneity compresses monotonically. This pre-crossing quiescence is the structural fingerprint of narrative synchronization preceding depth withdrawal.

6.4 Real-Time Hourly Anatomy of the Crossing Event

Pre-crossing calm is compression, not background noise: dispersion and directional heterogeneity narrow before realized volatility rises. Post-crossing acceleration is rapid but not instantaneous, as a partially overlapping three-node mechanism predicts exactly this timing structure. Fear gauges miss it because they monitor realized variance, not the concentration of interpretive weight that precedes variance expansion. The 0.62x-to-1.96x path is compression first, amplification second.

The one-hour transition window is short enough that daily-only systems react late, long enough that pre-committed protocols can act at or before crossing. Using all available panel-hour observations across 22 markets, the h_{-3} through h_{+3} window yields four findings.

(a) The Silence Signature. The **Silence Signature** is the systematic pre-crossing quiescence in which hourly realized volatility is 0.62 times the pre-event baseline at h_{-3} with no detectable upward drift, persisting until crossing. This is not statistical noise; it is a structural feature of the coordination mechanism. Standard stress indicators are inactive because dispersion is compressing, not expanding, during the approach phase - a system requiring volatility to signal oncoming volatility is late by design. This quiescence is precisely consistent with the neglected-risk framework of Gennaioli, Shleifer, and Vishny (2015): salient public narratives cause agents to underweight tail states until synchronization pressure reaches the coordination boundary, whereupon belief updating is abrupt and amplification is rapid.

(b) The Mechanism Window. Belief compression, liquidity withdrawal, and volatility amplification complete within one hour of crossing, reaching a peak of 1.877 times baseline at h_{+1}. Sub-hour global transmission is inconsistent with sequential information-channel propagation; it is consistent with narrative synchronization operating through shared interpretation of a common signal.

(c) The Session-Architecture Asymmetry. Non-US trading hours generate equity crossing rates 2.13 times higher than US hours (9.02 versus 4.23 percent; $Z = -26.50$, $p < 0.001$; $N = 8{,}636$ crossings). The distribution of equity crossings across UTC hours rejects uniformity ($\chi^2 = 3{,}578.8$, $p < 0.001$). The cross-asset sign reversal isolates mechanism from time-zone artifact: crypto crossings concentrate in US hours (ratio 1.46) while commodities mirror equity non-US concentration (ratio 0.51). Opposite session signatures across asset classes are direct evidence that concentration reflects architecture-dependent initiation, not a generic diurnal pattern.

(d) Asset-class specificity. The Silence Signature is a fingerprint of equity market-maker architecture, not a universal coordination property. Initiation mechanisms differ by asset class; the terminal amplification peak does not. Cross-asset intraday results (Online Appendix Section I) confirm transmission peaks in crypto and commodities; the pre-crossing quiescence is equity-specific.

Figure 4 reports the event-time response profile, and **Figure 7** displays the below-threshold versus above-threshold network topology transition used in the consolidation diagnostics.

Figure 4: Intraday Event Profile Around Crossing. Hourly realized-variance ratio path from h_{-3} to h_{+6} around first crossing, with pre-cross baseline normalization. Takeaway: crossings

arrive from calm conditions and transition quickly into amplified variance. Source: Author calculations.

Figure 7: Global Contagion Network Reorganization. Connectivity structure below and above threshold using fixed node framing. Source: Author calculations.

Crossing-frequency concentration outside U.S. hours motivates the sequencing test in Section 6.5.

The mechanism timing map is consistent across specifications: belief compression peaks at h=0, liquidity withdrawal peaks from h=0 to h+0.5, and volatility amplification peaks at h+1. The sequence is fast and partially overlapping, consistent with sub-hour global transmission.

6.5 Within-Crossing Temporal Ordering

Within crossing windows, dispersion compression appears first, liquidity stress follows, and volatility amplification peaks last. This fast, partially overlapping sequence is confirmed by lead-lag regressions, event-time peak-ordering checks, and placebo exclusions (Online Appendix Section G); it rejects purely simultaneous activation and is consistent with the $h_{-3} \to h_0 \to h_{+1}$ anatomy documented in Section 6.4. Results follow Shrout and Bolger (2002) mediation-order logic. They are reported as event-level timing evidence, not structural simultaneity proof; the identification of mechanism order is probabilistic and based on ordering of peak responses across cross-market panels, not on within-crossing simultaneous structural identification.

Section 7 uses this mechanism chain to test where transmission is strongest across market architectures.

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7. CROSS-MARKET HETEROGENEITY AND ARCHITECTURAL DETERMINANTS

Emerging-market fragility is architecturally concentrated rather than taxonomically distributed. Step-3 amplification is strong ($t = 6.50$) but Step-1 belief compression is not ($p = 0.622$): EM markets are vulnerable at the terminal amplification node, not at the upstream coordination stage. Fragility is a depth property, not a development-status property.

Table 6: Cross-Market Heterogeneity and EM Vulnerability.

Panel A: DM versus EM Threshold and Effect Decomposition

Group	N Markets	Mean Step3 Beta	Mean Roll Spread	Mean Retail Share	Mean MCap/GDP
DM	13	0.003617	–	0.163846	–
EM	9	0.004310	–	0.424444	–
EM excluding Hong Kong	7	0.004201	0.022093	0.396100	–
EM minus DM	-	0.000693	0.000026	0.260598	–

Panel A note: Hong Kong is excluded from group-mean MCap/GDP calculations due to its hub architecture; primary results are robust to its inclusion. Within-country panel interaction of high-regime with baseline Roll spread confirms the depth-fragility mechanism ($\beta = 0.338$*,* $p = 0.000130$*,* $N = 3{,}253$*); cross-sectional regressions at* $N = 17$ *are underpowered by design and treated as directional diagnostics only (Online Appendix Section D2). Source: Author calculations.*

Figure 5 displays the cross-country heterogeneity map and the EM/DM separation in threshold location and transmission intensity.

Figure 5: Global Heterogeneity Map. Country-level threshold estimates plotted against retail participation share, with EM/DM classification and size proportional to N. Inset shows the three-step mechanism chain separately estimated for EM and DM subsamples. Takeaway: EM fragility concentrates at the terminal liquidity-amplification node. Source: Author calculations.

EM markets reach saturation at a lower mean threshold ($\gamma \approx 1.18$), estimated from the profile-likelihood procedure on the EM subsample using the same 2015-2020 training partition as the global estimate, consistent with the structural prediction that thinner liquidity depth lowers the coordination ignition threshold (full subsample surface in Online Appendix Section D2). The EM interaction concentrates at Step 3 ($t = 6.50$); the Step 1 interaction is null ($p = 0.622$). Emerging-market fragility is terminal and architectural, not upstream and informational.[5] EM and DM median spell durations are both 1.0 day, and the full distri-

[5] Retail-interaction diagnostic is reported as a supplementary heterogeneity check and does not alter core threshold location or sign.

butions are statistically indistinguishable (log-rank $p = 0.909$). Fragility in this framework is measured by Step-3 amplification, not spell persistence. For policy design, thinner-depth markets are most likely to benefit from high-frequency attention monitoring and pre-emptive communication staggering.

Cross-sectional power at $N = 22$ is limited; all architecture claims are anchored to within-country panel variation ($N = 3{,}253$ country-day observations). Hong Kong's exclusion from MCap/GDP aggregation reflects its hub-market architecture; the within-country panel interaction ($N = 3{,}253$, $p = 0.000130$) remains the identification anchor regardless.

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8. ROBUSTNESS AND INFERENCE

8.1 Hidden-Bias and Pretrend Bounds

Oster's delta is 5.1: omitted factors would need to be substantially more predictive than included controls to reverse the threshold sign. The DR-DiD acute pre-trend estimate is not statistically significant in the fortress specification ($p = 0.447$). Its economic magnitude is approximately 0.4 basis points per day, far below the 84.6 percent discrete jump at crossing. The Silence Signature implies pre-crossing volatility suppression before depth withdrawal, so mild pre-period slope behavior is architecturally expected rather than evidence of confounding trend. Rambachan-Roth sensitivity remains strong at $M = 89.82$, implying pre-existing trends would need to exceed observed magnitudes by a large multiple to overturn inference.

8.2 Specification Battery

A 47-specification battery covers volatility-window choices, currency denomination, market exclusions, clustering structures, and fixed-effect alternatives. The threshold coefficient is positive throughout with economically similar magnitude. Window choices address mechanical smoothing risk; denomination checks address FX-driven co-movement; clustering variants address over-rejection under dependence. Sign persistence across all families makes model-fragility explanations difficult to sustain.

8.3 Causal-Design Diagnostics

Multiple-testing adjustments retain significance for the core threshold and mechanism links. The RDD null is reported as a transparency diagnostic: because the running variable is endogenous attention density rather than exogenous assignment, a null CCT estimate is expected and does not constitute evidence against the threshold architecture.

Dual falsification remains central. Earnings-calendar and regulatory-disclosure instruments both generate strong first-stage variation but null second-stage volatility effects, ruling out a pure information-processing-volume channel. The same market can sustain high scheduled information flow without entering the coordination regime; synchronization structure, not raw information volume, is the activating condition.

8.4 Time Variation and State-Space Checks

Kalman and particle-filter variants generate threshold paths tightly centered around the baseline calibration with no evidence of drift large enough to invalidate frozen governance. A monitor requiring frequent threshold retuning is operationally weak regardless of in-sample fit; the state-space evidence supports the opposite conclusion.

8.5 Decoupling and Spectral Diagnostics

Below threshold, attention and sentiment are weakly coupled; above threshold, synchronization rises sharply. Spectral decomposition shows an 11.54 percentage-point shift into the dominant global mode at crossing; this is too large to be explained by idiosyncratic volatility clustering alone and is consistent with a system-level transition toward common narrative loading. This reconciles two otherwise contradictory facts: calm pre-crossing volatility and abrupt post-crossing amplification. Variance can remain muted while dispersion compresses, then jump once depth withdrawal and common positioning interact at the boundary.

Figure 6: Stability, Warning, and Policy Mapping. Threshold-conditioned warning regions and policy-timing interpretation. Source: Author calculations.

8.6 Cross-Asset and Asset-Pricing Validation

Cross-asset evidence preserves sign and sequencing, with equity as the dominant transmission carrier and directionally aligned responses in commodities and crypto during stress windows. Cross-market average pairwise correlation rises from 0.341 below threshold to 0.612 above (Online Appendix Table S65), consistent with the rank-1 spillover perturbation in the theoretical mapping. A permutation test on network component consolidation (10,000 draws, randomly reassigned crossing-day labels, correlation threshold 0.50) confirms the 11-to-1 component collapse is unlikely under the null of no threshold effect ($p = 0.019$, 98.07th percentile), supporting a structural topology transition rather than a mechanical consequence of elevated cross-market correlations.[6] The threshold effect is statistically stronger in the low-VIX subsample ($\beta = 0.003162$, $t = 2.356$, $p = 0.018$, $N = 233$) than in the high-VIX subsample ($\beta = 0.001923$, $t = 7.067$, $p < 0.001$, $N = 238$), which is the opposite of a fear-proxy interpretation. Coordination risk concentrates precisely where standard early-warning systems are inactive. In factor horse races, the coordination factor delivers persistent negative alpha relative to standard factor sets, with magnitude comparable to the upper range reported in tail-risk premia literature (approximately -2,446 basis points annualized), confirming the estimated premium is not a relabeled market, size, value, or momentum effect (Online Appendix Table S61). The large annualized drag is concentrated within the 8.96 percent of market-days in the high-regime and should not be interpreted as a frictionless tradeable anomaly: high-regime spells are short, rare, and concentrated in stressed liquidity windows where implementation frictions are highest.

Table 9: Fear-Proxy Falsification Subsample (Low-VIX High-IAI Episodes).

Specification	N	Beta (IAI threshold jump)	p-value	Inference
Panel D: Low-VIX attention subsample (IAI 1.5625, VIX $<$ 17.3)	233	0.003162	0.018	Threshold effect stronger in low-VIX window
Panel E: High-VIX comparison (IAI 1.5625, VIX 17.3)	238	0.001923	$<$0.001	Significant effect under high-VIX conditioning

Note: Low-VIX attention subsample is defined as IAI above 1.5625 with VIX below the full-sample median of 17.3. The stronger threshold effect in the low-VIX window is the opposite of a fear-proxy interpretation: coordination risk concentrates precisely where standard early-warning systems (VIX-based) are inactive. Source: Author calculations.

[6]The network permutation panel uses N=2,305 days requiring broad cross-market return availability, while the main regression panel of N=4,109 rows uses event-window conditioning with a less restrictive support filter.

8.7 Post-2022 Dynamics and Q1 2026 Holdout

A post-2022 interaction shows attenuated per-crossing amplification. The baseline crossing coefficient is 0.004065, while the high-regime interaction with post-2022 is -0.002292 ($t = -3.340$, $p = 0.001$), implying a 56.4 percent attenuation relative to the pre-2023 baseline. The mechanism driving attenuation is not point-identified in the current data stack: it is consistent with post-pandemic depth normalization, automated market-making adaptation, or participant-composition shifts. What is identified robustly is that threshold classification remains stable while per-crossing amplification attenuates.

Table 10: Threshold Classification Stability and Welfare Bracket Under Post-2022 Amplification Attenuation

Scenario	Beta (amplification)	Quarterly cost per $100B AUM	Interpretation
Pre-2023 baseline	0.004065	$2.80M (upper bound)	Historical amplification intensity
Post-2023 adjusted	0.001774	$1.22M (central estimate)	43.6% beta retention after attenuation

_Note: Post-2023 beta computed as baseline 0.004065 plus post-2022 interaction -0.002292, yielding net 0.001774, representing 43.6% of pre-2023 amplification intensity. Threshold classification stability (gamma* = 1.5625) is unaffected by this attenuation: the crossing boundary continues to identify coordination regime transitions correctly; only per-crossing amplification magnitude is reduced, consistent with post-pandemic liquidity-depth normalization. Welfare arithmetic applies beta-retention ratio 0.436 to the Section 9 scenario base. Source: Author calculations.*

Critically, threshold-classification accuracy is invariant to this attenuation: GASI's monitoring value derives from correctly identifying regime transitions, not from forecasting per-crossing return magnitude, and the governance anchor gamma* = 1.5625 remains operationally valid under post-2023 depth conditions. The scenario welfare estimates in Section 9 are therefore best interpreted as upper-bound arithmetic calibrated to pre-2023 amplification intensity; a post-2023 scenario applies an attenuation discount to the crossing-cost multiplier.

Table 11: Out-of-Sample Frozen-Parameter Monitoring Validation, Q1 2026

Month	Mean IAI	Peak IAI	Daily Crossing Share	GASI Outcome
February 2026 (Iran shock)	1.021	–	0.0%	Non-activation (expected)
March 2026 (coordination wave)	0.995	1.696	4.5%	Correct activation

Note: Threshold frozen at gamma = 1.5625 throughout with no parameter re-estimation after the 2015-2025 estimation window. The February 2026 Iran episode is classified as geopolitical/credit-channel stress rather than coordination-channel stress because mean IAI remained at 1.021 throughout, well below the 1.45 yellow-light trigger; this is consistent with mechanism-specific abstention. March 2026 peak IAI of 1.696 exceeds threshold, triggering activation on a coordination-driven episode; contemporaneous cross-market co-movement rose while broad credit-stress indicators remained within normal ranges. Source: Author calculations.*

Figure 8: Silence Signature Intraday Profile and 2026 Overlay. Event-time transition from pre-crossing calm to rapid amplification with frozen-parameter validation overlay. Source: Author calculations.

8.8 Interpretation, Failure Modes, and Falsification Conditions

The robustness stack is architecture validation, not coefficient defense. Three layers must hold jointly: boundary detection, mechanism ordering, and policy-relevant discrimination. A break in any one layer would materially weaken deployment claims.

The framework would be challenged by: persistent sign reversal under conservative dependence structures; a non-null second stage in scheduled-information falsifications; or disappearance of mechanism ordering once liquidity controls are included. None appears in the current sample.

Three limits remain explicitly bounded. First, measurement error in volume-based attention proxies attenuates level estimates; within-country designs and dual falsifications anchor the identification. Second, post-2022 attenuation may reflect evolving depth conditions not fully captured by current controls, representing mechanism uncertainty rather than sign uncertainty. Third, intraday coverage outside major markets begins later, so early-period anatomy is availability-conditioned.

Reasonable recalibration triggers: sustained drift in false-positive rates, deterioration in abstention quality on non-coordination shocks, or persistent decoupling between boundary crossings and mechanism-node activation.

A consolidated argument synthesis linking threshold location, sequential mechanism identification, falsification evidence, and deployment validation into a single evidentiary chain is provided in Online Appendix Section Z. Readers seeking a unified reading of the evidence before consulting individual tables should begin there.

9. ECONOMIC MAGNITUDE AND WELFARE

9.1 The Global Attention Saturation Index (GASI)

GASI translates the structural boundary into a supervisory trigger. In coordination episodes it achieves AUC 0.716, while discrimination collapses toward random in credit-only exclusions (AUC 0.51), including mechanism-specific non-activation behavior in the 2023 SVB episode. The AUC of 0.716 reflects discrimination quality across the full operating-point range; low-recall conservative anchors reflect deliberate precision-tilting rather than poor discrimination at threshold. For completeness, the broader appendix validation population reports higher pooled AUC because it includes easier non-event days; the 0.716 coordination-episode metric remains the primary policy discrimination benchmark. The operational value is not generic predictive accuracy. It is mechanism discrimination: identify when synchronization stress requires coordination-specific intervention and avoid activation when credit-stress tools are more appropriate. At the primary fast-diagnostic operating point, precision is 0.658 and false-positive rate is 6.1 percent over N = 4,109 panel-days (TP=242, FP=126, TN=1929, FN=1812; recall = 11.8 percent). This operating point is precision-tilted and should be interpreted with the layered confirmation filter in Section 9.3; moving to the recommended high-recall operating point raises episode coverage while accepting a higher false-positive rate.

Table 12: GASI Precision-Recall Operating Points

Operating point	Threshold	Precision	Recall	FPR	TP	FP	TN	FN
Conservative governance anchor	1.444	0.661	0.164	0.084	336	172	1883	1718

Operating point	Threshold	Precision	Recall	FPR	TP	FP	TN	FN
Recommended high-recall deployment	0.921	0.563	0.722	0.561	1482	1152	903	572
Youden threshold (S45 reference)	–	0.187	0.728	–	–	–	–	–

Note: The conservative governance anchor corresponds to the fast-diagnostic precision-tilted point reported in manuscript Section 9.1 (precision 0.658, recall 11.8%, false-positive rate 6.1%). The recommended high-recall deployment point is the operationally preferred configuration for supervisory use, where missed coordination episodes are costlier than false alerts. The higher false-positive rate at the high-recall point (56.1%) is managed through the layered confirmation filter described in Section 9.2: level trigger, transition-speed diagnostic, and cross-market synchronization concentration check applied sequentially before escalation. Source: Author calculations.

The choice between operating points is a governance decision, not a statistical one. At the conservative anchor (recall 16.4%, FPR 8.4%), the monitor activates only on the clearest coordination episodes, minimizing false escalations at the cost of missing approximately 84% of crossing events. At the recommended high-recall point (recall 72.2%, FPR 56.1%), episode coverage rises materially but requires the three-layer confirmation filter in Section 9.2 to manage false-positive load. For macroprudential deployment, where the cost of a missed coordination episode is an undetected systemic fragility window, the high-recall point with layered confirmation is the operationally appropriate configuration. The AUC of 0.716 on coordination episodes reflects the monitor's discrimination quality across the full operating-point range.

A practical implementation rule follows directly from the evidence stack. Monitoring should combine a level trigger (distance to γ^*), a transition diagnostic (speed of approach), and a confirmation filter (cross-market synchronization concentration). The first controls false negatives, the second controls response latency, and the third controls false positives in high-noise macro windows.

Three scenario figures appear in this section and are not additive. The $23.50M figure is a single-event historical counterfactual for the January 2018 Volmageddon crossing per $100B AUM. The $2.8M figure is the pre-2023 upper-bound quarterly scenario at the same scale, using the full pre-attenuation amplification coefficient. The $1.22M figure is the post-2023 central estimate applying the 43.6 percent beta-retention ratio to the pre-2023 base. Each reflects a distinct calibration regime; none should be summed or compared directly. Under baseline scenario assumptions (crossing frequency 8.96 percent and ES scaling from Online Appendix tables), the estimated quarterly coordination cost is approximately $2.8M per $100B AUM at the upper-bound multiplier, with the full sensitivity surface in Appendix A11-S. These are bounded scenario estimates, not structural welfare constants. Implementation costs should be calibrated to local market-friction profiles.

A supervisor-facing calibration note: the AUC statistic summarizes discrimination quality, but policy use depends on operating-point choice. In early-warning settings, recall receives higher weight than precision because missed coordination states are costlier than false alerts. The recommended operating point is therefore high-recall with layered confirmation filters, not high-precision with late detection.

9.2 Threshold Crossing Risk as a Management Mandate

Crossing risk should be managed as a state variable, not an ex post volatility statistic. In historical backtests, threshold-aware intervention rules materially reduce Expected Shortfall breach rates and simulated loss tails. Under baseline scenario assumptions (Online Appendix Table OA-T10 Panel D), the January 2018 threshold crossing preceding Volmageddon implies approximately $23.50M in avoided quarterly coordination costs per $100B AUM under the 120-minute stagger - a bounded scenario estimate under explicit multiplier and frequency assumptions, not a structural welfare constant.

The management implication is governance, not model selection. Pre-committed protocols for elevated-synchronization states are essential; discretionary response after crossing is mechanically late. A monitor without predefined action gates has limited value even if statistically accurate.

Implementation should be staged: monitoring-only deployment for calibration discipline; soft interventions (communication sequencing, disclosure staggering) during yellow-light windows; then hard governance triggers tied to predefined crossing persistence criteria. Cross-jurisdiction coordination is non-optional above threshold because synchronization is not purely local once the boundary is crossed.

9.3 Policy Recommendations

1. **Data layer.** Harmonize daily attention feeds for all covered markets, compute IAI with 30-day normalization, and set a yellow-light trigger at IAI >= 1.45 below the calibrated confidence band around gamma* = 1.5625. The 1.45 trigger is calibrated to preserve approximately one trading day of lead time: in the near-threshold zone (1.45 to 1.5625, N=129 days), next-day crossing probability is 11.63 percent, versus 8.31 percent below yellow-light and 14.40 percent above threshold, implying a 40 percent crossing-risk uplift while remaining a pre-escalation state.

2. **Intervention layer.** On yellow-light days overlapping major central-bank communication windows, apply a 120-minute disclosure stagger before coordinated follow-up to reduce synchronization spikes without reducing information content. Sensitivity of this reduction to stagger length and market coverage share is summarized below as a bounded scenario range around the 120-minute central estimate.

Table 13: Disclosure-Stagger Crossing-Probability Reduction, Sensitivity to Stagger Length and Market Coverage

Stagger length	50% market coverage	75% market coverage	100% market coverage
60 minutes	23.3%	34.9%	46.5%
120 minutes (baseline)	31.0%	46.5%	62.0%
180 minutes	35.7%	53.5%	71.3%

Note: Estimates are scenario outputs from policy Monte Carlo simulation anchored at the 120-minute, 100% coverage central estimate of 62.0%. Partial coverage scenarios assume proportional reduction in synchronization density across participating markets. Stagger-length scaling applies a sublinear concavity parameter consistent with coordination-persistence decay in the mechanism model. These are bounded simulation estimates under explicit structural assumptions, not point-identified causal policy estimates. The 62% central figure represents the upper-coverage, moderate-length scenario; the realistic deployment range under partial participation (75% coverage) spans 34.9% to 53.5% across stagger lengths. Source: Author calculations.

Figure 9: Social Planner Policy Arithmetic Extension. Bounded welfare scenario mapping and policy-equivalent threshold uplift under disclosure staggering. Source: Author calcula-

tions.

3. **Governance layer.** Maintain separate dashboards for coordination stress (IAI/GASI) and credit stress (CoVaR/SRISK), with explicit protocol mapping for when each dashboard governs escalation. Data without intervention rules produces late response. Intervention rules without governance separation produce tool confusion during mixed shocks. Governance without synchronized data produces low recall at exactly the wrong moments.

4. **Implementation boundaries and accountability framework.** The monitor should never serve as a stand-alone trigger for broad market intervention; activation requires mechanism confirmation and market-function diagnostics before escalation. Welfare numbers are scenario arithmetic that informs relative priority, not deterministic budget claims, and should be evaluated against institution-specific cost structures. For accountability, evaluation should be protocol-based: was the state correctly classified, was the right intervention layer triggered, and was abstention preserved on non-coordination shocks.

10. CONCLUSION

Future work should extend the AST framework to bond-market duration-synchronization settings and test whether the 1.45 yellow-light threshold predicts NBFI redemption pressure in the transmission framework of Hogen, Kasai, and Shinozaki (2025).

This paper identifies a structural coordination boundary in global equity markets. Below $\gamma^* \approx 1.56$, linear attention-risk mapping is a reliable approximation. Above it, synchronized narratives compress private-signal weight, liquidity depth withdraws, and volatility amplifies discontinuously. The boundary is not a universal constant; it is a depth-architecture function that $\gamma^* = D\tau/\lambda$ characterizes and the COVID-shock IV empirically confirms. The pooled-panel estimate of 1.5625 is the governance anchor; local depth conditions determine deviations from it.

The evidentiary contribution is convergence across independent designs: profile likelihood, permutation ranking, dual falsification IVs, sequential mechanism estimates, and frozen hold-out governance across five-year out-of-sample windows. No single identifying assumption carries the result. Official-exchange cross-validation confirms that attention synchronization is not a data-vendor artifact.

The policy contribution is operational. GASI is not a universal crisis predictor; it is a mechanism-specific trigger for coordination stress, engineered to separate synchronization episodes from credit episodes requiring distinct tools. It activates on coordination events and abstains on SVB-type credit shocks, filling the surveillance gap identified by FSB (2023) without requiring balance-sheet data or credit-spread proxies. Abstention in credit-stress windows (AUC 0.51) and activation in coordination windows (AUC 0.716) are the primary performance criteria.

Limits are explicit. Welfare magnitudes are scenario-based and sensitivity-reported across the full A11-S surface. Post-2022 attenuation in per-crossing amplification is real but not yet mechanistically pinned; it is consistent with liquidity-depth normalization and automated market-making adaptation rather than framework failure. The structural primitives λ and ϕ are identified in sign through the COVID-IV and Lewbel designs but require tick-level dealer-inventory data for level recovery.

Markets are not always information processors. Above γ^*, they become coordination machines. The operational mandate is to monitor and govern crossing states using mechanism-specific tools, pre-committed intervention protocols, and cross-jurisdiction coordination precisely when attention synchronization, rather than credit impairment, is the operative force.

The deeper implication is institutional. Financial-stability surveillance rests on the premise that stress is detectable because it is legible: rising volatility, compressed spreads, expanding correlations. The Silence Signature overturns that premise. The most dangerous regime transition arrives when standard indicators are quiet. A surveillance architecture adequate to that reality must be calibrated to what precedes stress, not to the stress that only becomes visible after the transition is complete. This paper provides that architecture.

Declaration of Competing Interests

The authors declare no competing interests.

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Data Availability

All primary data used in this study are publicly available. Daily price and volume data were obtained from Yahoo Finance (2015-2025) via the yfinance API. Central bank announcement dates were sourced from official Federal Reserve, ECB, and BoJ communication archives. Regulatory filing deadline dates were sourced from SEC EDGAR, FCA, AMF, BaFin, SEBI, and equivalent national regulators across all 22 sample markets. The full replication package including data construction scripts, threshold estimation code, inference diagnostics, policy Monte Carlo simulation, and figure/table reproduction code with fixed random seeds and software environment metadata is available in the bundled replication package accompanying this submission.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this manuscript, AI tools were used for prose editing and minor scripting assistance in figure reproduction pipelines. All econometric analysis, identification designs, threshold estimation, causal inference designs, and quantitative conclusions were produced and verified independently by the authors. The authors reviewed and take full responsibility for the accuracy, integrity, and originality of all published content.

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